

ASSESSMENT BRIEF

Programme:	MSc Data Science		
Module Title	Data Visualisation		
Module Code	CMP020L013		
Module Level	7	Assessment Type(s)	Coursework
Word Length / Duration	4000	% contribution to module mark	100%
Deadline (date & time) for Submission	24 th April 2026	Format/Location of submission	Moodle
Assessment Feedback date:	1 st May 2026		
Learning Outcomes LO1: Critically analyse the effectiveness of a given visualisation technique for a particular task. LO2: Critically evaluate the appropriate visualisation technique for specific problems. LO3: Develop visualisations to report on studies of both quantitative and qualitative evaluations.		Employability and Professional Skills This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the following skill(s) which is(are) critical in professional context and jobs. By successfully completing this assessment, you will be able to ... • Ability to create clear and effective data visualisations to support decision-making. • Strong analytical and critical evaluation skills in selecting appropriate visualisation techniques. • Improved communication skills through visual storytelling for non-technical audiences.	

	• Enhanced technical skills in Python and data analysis tools. • Experience applying data-driven insights to real-world business problems.
Assessment Requirements Data Visualisation Assessment Brief (CMP020L013) <i>The summative assessment for this module comprises two mandatory components: (1) an artefact (Python code implementation) and (2) a written report, collectively accounting for 100% of the final mark.</i> 1. Introduction Understanding customer purchasing behaviour and product performance is crucial for organisations aiming to optimise sales strategies, improve customer engagement, and enhance profitability. Businesses rely on data-driven insights to make informed decisions about product offerings, pricing, marketing, and customer retention. Analysing sales transactions at the customer level enables businesses to identify high-value segments, track purchasing patterns, and assess the effectiveness of sales interventions. In this context, a comprehensive sales dataset can support strategic decisions across departments, including marketing, operations, and finance. 2. Dataset Information The <i>Prodcut_Sales.csv</i> dataset provides a detailed snapshot of product sales at the customer level. It includes records of individual transactions, capturing key attributes such as customer ID, product ID, quantity sold, unit price, date of purchase, and product category. The dataset also includes customer-specific features that can support segmentation and targeted analysis. This data enables detailed exploration of customer-product interactions and supports use cases such as market basket analysis, sales forecasting, and customer lifetime value estimation. The metadata of the dataset is available on the Moodle page (<i>Metadata_Prodcut_Sales.txt</i>). 3. Tasks 3.1 Data Preparation and Exploratory Data Analysis (EDA) (20 Marks) • Clean and preprocess the dataset by handling missing values and outliers • Perform statistical analysis to understand data distribution • Document all data cleaning steps and transformations 3.2 Research Questions and Integration of Domain Knowledge (30 Marks) • Formulate three specific research questions based on the dataset	

- Integrate domain-specific knowledge to formulate your questions
- Ensure questions are answerable with available data
- Justify the relevance and importance of each question

3.3 Data Visualisation (40 Marks)

- Create appropriate visualisations relevant to previous tasks and research questions
- Apply appropriate colour schemes, visual encoding, labels, and legends
- Optimise visualisations for clarity, impact taking into account your potential audience

3.4 Report Compilation (10 Marks)

Compile a 4,000-word report (excluding references and appendices) using the provided template on Moodle (*Report Submission Template*) that synthesises all aspects of your analysis and visualisations into a cohesive narrative. The report should cover:

- Detailed interpretation of each visualisation
- Identify key patterns, trends, and relationships
- Support findings with statistical evidence
- Draw meaningful conclusions from the visualisations
- Address potential limitations and biases
- Structure report with clear sections and flow
- Include proper citations following the IEEE format
- Create an engaging narrative connecting visualisations
- Provide a summary of findings and limitations

4. Deliverables

The assessment requires the mandatory submission of two components:

1. **Artefact:** Python script (.py) or Jupyter notebook (.ipynb), named with your student number (*ABC12345678.py/.ipynb*)
2. Report: A written report named with your student number, preferably in PDF format (*ABC12345678.pdf*)

- Use the provided report template for your

4.1 Technical Requirements

- The Python implementation file (.py/.ipynb) and the dataset (*Prodcut_Sales.csv*) must be collocated in the working directory (same folder) to ensure proper execution. This single-directory configuration is mandatory for the assessment.
- The Python code must successfully load the dataset from the working directory on university machines running a Python 3.10+ runtime environment and

generate the reported output without issues as evidence of analysis and visualisations; otherwise, you will fail the assessment. - You must submit both parts for a valid submission; otherwise, you will fail the assessment. - While coding quality and style are not directly assessed, clear and well-commented code is expected.					
Marking Criteria					
Criteria	0-49% (Fail)	50-59% (Pass)	60-69%	70-79%	80-100%
Data Preparation and Exploratory Data Analysis (EDA) (20 Marks)	Minimal or ineffective data cleaning; critical issues remain unaddressed, leading to unreliable analysis. Lacks critical analysis or insight; limited or superficial examination of data characteristics. Python code fails to load the dataset from the working directory; data cleaning/EDA calculations fail to execute; the reported visualisations cannot be generated.	Basic data cleaning attempted, addressing some issues but leaving gaps; limited rigor in handling outliers. Basic EDA with limited insights; lacks depth in identifying patterns, trends, or relationships. Python code successfully loads the dataset from the working directory; most data cleaning/EDA calculations are executed, and only some of the visualisations are generated successfully.	Adequate data preparation with minor issues; most missing values and inconsistencies addressed. Competent EDA showing good understanding of data characteristics, with some insightful observations. Python code successfully loads the dataset from the working directory; most data cleaning/EDA calculations are executed, and most visualisations are generated as reported.	Comprehensive data cleaning with minimal adjustments needed; data is thoroughly prepared for analysis. Detailed EDA with strong critical analysis; insightful observations on trends, distributions, and relationships. Python code successfully loads the dataset from the working directory; all reported data cleaning/EDA calculations are executed, and all reported visualisations are generated as reported.	Exemplary data preparation with rigorous handling of all issues; data is fully ready for robust analysis. Advanced EDA demonstrating exceptional critical insight; provides in-depth understanding of complex patterns, trends, and relationships. Python code successfully loads the dataset from the working directory; all reported data cleaning/EDA calculations are executed and well-commented, and all reported visualisations are generated as reported.

Research Questions and Integration of Domain Knowledge (30 Marks)	Absent or irrelevant; lacks focus or relevance, providing minimal guidance for analysis; no integration of domain knowledge; minimal contextualization.	Basic questions that are somewhat relevant but lack critical depth and focus; minimal integration of domain knowledge	Relevant and clear research questions that guide meaningful exploration, showing moderate insight, and relevant domain knowledge but lack depth.	Well-developed questions that drive a focused, critical exploration of the data, revealing clear insights, domain knowledge is well-integrated to enhance understanding.	Highly insightful, relevant, and in-depth questions that guide complex analysis; demonstrate critical understanding, expertly integration of domain knowledge, and current research, adding significant context and depth.
Data Visualisation (40 Marks)	Visualisations are poorly designed, unclear, or fail to support the analysis meaningfully.	Some visualisations present but lack clarity, relevance, or impact; limited support for analysis; limited application of visual encoding and colour.	Clear and mostly relevant visualisations that support analysis; Effective visualisations that are mostly clear and relevant; appropriate use of visual encoding and colour	well-crafted visualisations that enhance clarity and depth of analysis. Highly effective visualisations; well-designed application of visual encoding and colour for clarity and insight.	Expertly designed visualisations that are clear, highly relevant, and demonstrate critical insight; add significant depth to the analysis. Exceptional visuals are expertly crafted with excellent visual encoding and colour that enhance insight and clarity.

Structure, Writing Style, Storytelling with Data and Referencing (10 Marks)	Poor organisation, lacking logical flow; frequent language errors; lacks a coherent narrative, missing or incorrect citations; and inconsistent IEEE referencing.	Basic structure with limited coherence; some language issues; basic narrative with limited insights; citations used but with IEEE referencing errors.	Well-organized report with clear explanations; logical flow; mostly, narrative conveys some insights; accurate IEEE referencing with minor citation issues.	Strong structure and clear, precise language; strong narrative with critical insights; accurate citations; IEEE referencing used consistently.	Exceptionally organised, cohesive, and engaging; precise language; compelling, well-rounded narrative demonstrating exceptional understanding; critical insight and in-depth, thorough and accurate use of IEEE referencing; citations well-integrated.
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Use of Artificial Intelligence (AI)

In this section include details of the use of AI tools to support the assessment.

Please read and apply:

<https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4a%20AI%20principles%20and%20guidance%20-for%20Senate.docx>

<https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4b%20Student%20Guidelines%20on%20the%20use%20of%20AI .docx>

The assessment is designed so that the use of AI during the assessment is possible. You must acknowledge any use of AI and appropriately cite all AI generated outputs. Please make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions. You can find the guidance on the use of AI.

Referencing

IEEE style: <https://www.citethemrighttutorial.com/course-modules/ieee>

Mitigating circumstances/late penalties

Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton

Student Union (RSU) or someone in the [Wellbeing team](#) first, who can support you. Further details can be found on the [mitigating circumstances portal](#).

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 50%; if it is over 14 days late, it will not be marked.

Submission Checklist

Before you submit, ask yourself:

- Have I fully answered the assessment brief?
- Have I met the word count and formatting requirements?
- Is my referencing complete and accurate?
- Have I declared any AI use honestly?
- Have I proofread my work?
- Am I submitting through the correct platform before the deadline?